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Multi-scale modeling approaches in wire arc additive manufacturing: from melt pool dynamics to residual stress prediction

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Abstract

Wire Arc Additive Manufacturing (WAAM), a subset of Directed Energy Deposition (DED), is gaining prominence for producing large-scale metallic components in aerospace, defense, and marine sectors. Despite its cost-effectiveness and scalability, WAAM faces challenges related to thermal gradients, residual stresses, distortion, and microstructural heterogeneities. To overcome these issues, multi-scale modeling has become a powerful tool for process understanding, optimization, and prediction. This chapter explores the integration of finite element analysis (FEA), thermal–mechanical simulations, and phase-field modeling techniques to predict and control WAAM-induced phenomena across multiple spatial and temporal scales. At the macro-scale, finite element and thermal–mechanical models are employed to analyze the global heat flow, temperature distribution, and stress evolution during deposition. These models help predict deformation, residual stresses, and interlayer bonding, which are essential for ensuring structural integrity. At the mesoscale, simulations capture melt pool geometry, thermal cycles, and fluid dynamics, enabling insights into bead shape and dilution characteristics. Finally, at the microscale, phase-field models are used to simulate grain growth, microstructural transitions, and solidification kinetics, all of which directly influence mechanical properties. This chapter also details the coupling strategies among the thermal, mechanical, and metallurgical domains, along with validation using experimental data such as thermocouple readings, X-ray diffraction (XRD) for residual stress, and EBSD for grain morphology. Advanced modeling approaches like cellular automata, level-set methods, and reduced-order models are discussed to enhance computational efficiency and accuracy. Additionally, the use of digital twins and data-driven models (e.g., neural networks trained on simulation output) is explored to provide real-time feedback and predictive capabilities. The chapter concludes with case studies demonstrating the successful implementation of multi-scale models to optimize WAAM parameters, reduce defects, and improve dimensional accuracy. By presenting a holistic modeling framework, this chapter offers engineers, researchers, and practitioners a roadmap for integrating simulation-driven design and control strategies into WAAM systems to improve performance, reduce trial-and-error, and enable sustainable manufacturing.

Extended author information available on the last page of the article

Keywords Wire arc additive manufacturing (WAAM) · Multi-scale modeling · Thermo-mechanical simulation · Melt pool dynamics · Microstructure evolution · Digital twin

1 Introduction to WAAM and the need for modeling

Wire Arc Additive Manufacturing (WAAM) is an advanced direct energy deposition technique that leverages arc welding processes to build large-scale metal components layer by layer. Initially introduced through methods like three-dimensional welding [1, 2], WAAM has emerged as a viable option for industrial sectors requiring customized, structurally sound parts. Unlike traditional subtractive machining, WAAM reduces material waste and production time [3], making it a promising solution for cost-effective manufacturing [4]. However, challenges such as thermal distortion, residual stress accumulation, and poor surface finish remain central concerns, necessitating the integration of computational tools for modeling and prediction [5]. The process, when applied to materials such as Ti–6Al–4 V [6, 7], requires precise control over arc parameters to ensure mechanical integrity and microstructural consistency.

Predictive modeling is critical in this context to preemptively assess heat accumulation, mechanical stresses, and geometric deviations during layer deposition. WAAM involves complex thermal cycles that can lead to significant mechanical deformation [8], requiring simulation frameworks to manage design and manufacturing accuracy. Directed energy deposition studies [9, 10] have highlighted how real-time simulation can assist in parameter tuning, thereby reducing trial-and-error experimentation. Simulation-driven manufacturing also minimizes resource usage, aligning with sustainable manufacturing goals [11]. Moreover, integrating open-source platforms and sensors into WAAM has accelerated the development of digital twins and closed-loop control systems [12, 13]. Such advancements underscore the growing emphasis on real-time feedback and adaptive modeling, which are pivotal for overcoming inconsistencies such as bead profile irregularities or geometric warping [14]. Combining modeling with inter-pass techniques, such as rolling [15], has also been shown to significantly influence the resulting microstructure and reduce porosity, reinforcing the need for simulation to achieve optimal quality.

Hybrid approaches—combining WAAM with milling or laser processing—introduce additional complexity to process planning and necessitate multiphysics simulation for hybrid system validation [16, 17]. Furthermore, new developments such as dual-wire plasma arc deposition [18] or pulsed-current-based shaped metal deposition [19] reveal the widening scope of WAAM, in which simulation plays a key role in understanding interaction effects and optimizing process parameters. Advanced process monitoring systems [20] also rely on real-time simulation to improve layer homogeneity and control thermal gradients, which are known contributors to anisotropic mechanical behavior [21]. Overall, the evolution of WAAM from a rapid prototyping too [22] to a full-fledged manufacturing process [23] highlights the central role of simulation and modeling in supporting its industrial adoption. Whether evaluating the impact of arc behavior on component geometry [24] or estimating process outcomes such as deposition accuracy and build time [25], predictive modeling emerges as a linchpin for ensuring repeatable and scalable production. Simulation tools offer manufacturers a digital sandbox to optimize parameters such as wire feed

rate, arc power, and deposition speed without physical iteration, promoting innovation and quality assurance. With continued research and integration into digital manufacturing ecosystems, modeling in WAAM stands to revolutionize the way large-scale metallic parts are conceptualized and manufactured.

2 Macro-scale thermal and mechanical modeling

Macro-scale modeling in Wire Arc Additive Manufacturing (WAAM) focuses on capturing the broader thermal and mechanical behavior during layer-by-layer deposition. Given the inherently high heat input of arc-based deposition techniques, thermal management is a primary concern because it directly influences residual stress formation, distortion, and metallurgical properties [23]. Finite element methods (FEM) have become the dominant approach for macroscale simulations, enabling the estimation of temperature fields, heat-flow paths, and mechanical deformation in three dimensions [24]. For instance, Hamilton et al. [21] employed a thermal-mechanical model to simulate the WAAM process. They revealed how heat accumulation across successive layers causes non-uniform expansion and contraction, leading to stress gradients and deformation. Such simulations are critical for high-integrity components in aerospace and defense sectors, where dimensional accuracy and mechanical reliability are paramount.

One of the earliest contributions to macro-scale thermal modeling was provided by Ribeiro et al. [22], who evaluated heat distribution during WAAM and its effect on build geometry. Their findings were instrumental in understanding that interlayer cooling time and arc travel speed significantly influence the thermal history of the build. Subsequent studies by Zhang et al. [20] expanded on this by incorporating real-time thermal sensors into their simulation setups to validate the FEM models and bridge the gap between theoretical predictions and experimental data. More recent work by Hu et al. [14] integrated process monitoring with predictive modeling to improve the accuracy of heat distribution profiles, enabling better planning of interpass temperature control.

Mechanical stress prediction is often coupled with thermal simulations to understand how variations in heat inputs and cooling rates affect residual stress and deformation. Corbin et al. [8] demonstrated a coupled thermal-mechanical model for Inconel 718 in WAAM, showing how thermal gradients across the height of the part lead to uneven shrinkage and tensile residual stress zones. This was further supported by Li et al. [11], who investigated stress evolution in aluminum alloys and found that optimizing arc parameters and deposition strategies could reduce warping by over 30%. Such results are crucial when scaling WAAM for larger components, as thermal accumulation becomes more pronounced with increased part volume and layer count. Macro-scale models also help identify critical process windows to avoid defects such as delamination or excessive bulging. Sequeira Almeida and Williams [6] used FEM to simulate the deposition of Ti–6Al–4 V and confirmed that preheating the substrate combined with controlled cooling could minimize stress concentrations. Similarly, Thomas and Gilbert [3] demonstrated that controlling the dwell time between successive layers is essential to prevent interlayer cracking due to rapid temperature cycling. These insights support the development of closed-loop control systems and automated process adjustments, further bridging the simulation-to-fabrication gap.

2.1 Heat conduction equation

$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q$$

where

- ρ = density (kg m^{-3}).
- C_p = specific heat capacity (J kg K^{-1}).
- T = temperature (K).
- k = thermal conductivity (W m^{-1} K^{-1}).
- Q = internal heat generation (W m^{-3}).

The transient heat conduction equation governs the evolution of temperature during WAAM and forms the basis of finite element thermal simulations used to predict thermal history and cooling rates.

2.1.1 Mechanical equilibrium equation

$$\nabla \cdot \sigma + b = 0$$

where

- σ = stress tensor.
- b = body force.

The mechanical equilibrium equation is solved together with constitutive relations to predict residual stress and distortion resulting from thermal expansion and contraction during deposition.

Beyond thermal and stress modeling, macro-scale simulations are increasingly used to predict shape deformation and distortion in real time. Xu et al. [12] developed a digital twin model for WAAM that integrates temperature, displacement, and stress data to forecast the component's final shape. Such predictive models have immense value in high-precision industries, especially when used in conjunction with adaptive slicing and deposition planning algorithms. Feng et al. [18] demonstrated that combining dual-wire plasma arc deposition with thermal simulations can enable near-net-shape production of large structural components, reducing the need for extensive post-processing. Moreover, hybrid simulation strategies that combine macro-thermal analysis with path optimization algorithms are now being explored. Shi et al. [16] introduced a toolpath-based thermal analysis to optimize deposition order and minimize heat-affected zones. Their approach reduced peak thermal accumulation by up to 18%, showcasing how intelligent simulation can enhance both part quality and energy efficiency. McAndrew et al. [15] expanded on this by incorporating interpass rolling into their models, showing that it not only alters thermal profiles but also homogenizes mechanical stress distribution, enabling stronger and more uniform

builds. Overall, macro-scale thermal and mechanical modeling is indispensable for achieving repeatability, reliability, and precision in WAAM. By simulating temperature gradients, stress propagation, and structural deformation, these models enable proactive tuning of process parameters and deposition strategies. With advancements in computational power and real-time data acquisition, macro-scale models are increasingly transitioning from offline planning tools to integral components of live manufacturing systems, paving the way for smarter, more autonomous WAAM platforms.

Figure 1 presents the hierarchical multi-scale modeling framework employed in Wire Arc Additive Manufacturing (WAAM). The process begins with the selection of key process parameters, including current, voltage, wire feed rate, and travel speed, which determine the heat input and material deposition characteristics. These inputs influence three interconnected modeling scales. Macro-scale modeling predicts thermal history, temperature distribution, residual stress, and distortion. Meso-scale modeling captures melt pool dynamics, fluid flow, heat transfer, and solidification behavior. Micro-scale modeling simulates grain growth, phase transformation, and microstructure evolution. Integrating these modeling scales enables accurate prediction of final mechanical properties, including strength, hardness, fatigue performance, and dimensional accuracy in WAAM-fabricated components.

Table 1 provides a comparative overview of the principal multi-scale modeling approaches employed in Wire Arc Additive Manufacturing (WAAM), highlighting their objectives, modeling techniques, key outputs, advantages, and limitations. Macro-scale modeling primarily focuses on predicting thermal history, residual stress, and component

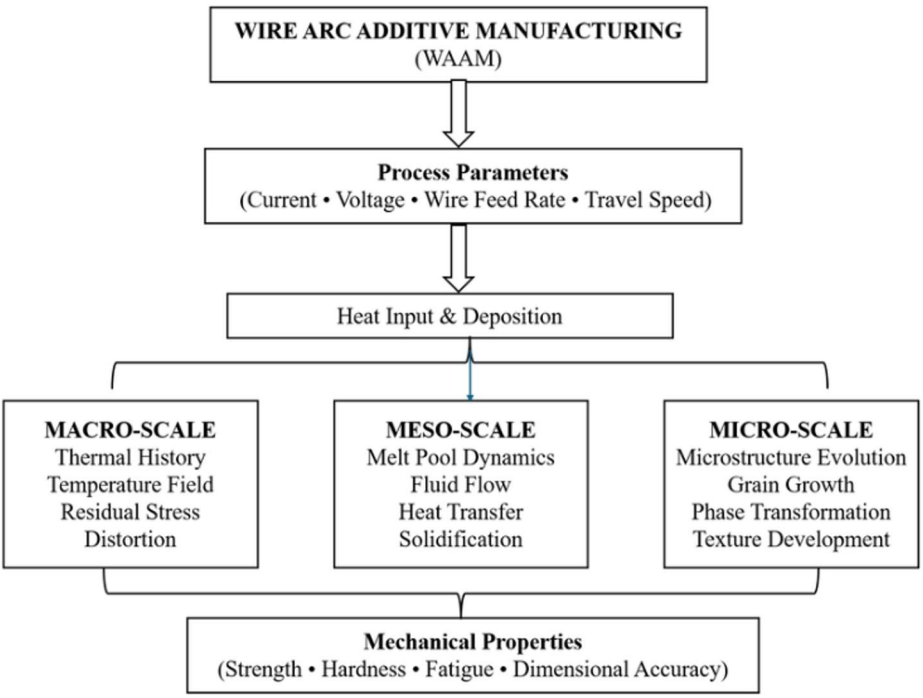


Fig. 1 Multi-scale modeling framework in wire arc additive manufacturing (WAAM), illustrating the interaction between macro-, meso-, and microscale phenomena

Table 1 Comparison of multi-scale modeling approaches in wire arc additive manufacturing (WAAM)

Modeling Scale	Primary Objective	Modeling Techniques	Key Outputs	Advantages	Limitations
Macro-scale	Predict thermal and mechanical behavior	Finite Element Analysis (FEA), Thermo-mechanical models	Temperature distribution, residual stress, distortion	Efficient for component-level analysis and process optimization	Limited capability to capture melt pool and microstructural phenomena
Meso-scale	Analyze melt pool behavior and solidification	Computational Fluid Dynamics (CFD), Volume of Fluid (VOF)	Melt pool geometry, fluid flow, heat transfer, solidification	Provides detailed understanding of melt pool dynamics	Computationally intensive with high mesh requirements
Micro-scale	Predict microstructure evolution	Phase-field modeling, Cellular Automata (CA), Crystal Plasticity	Grain growth, phase transformation, texture evolution	Enables prediction of material properties and microstructural characteristics	High computational cost and requires detailed material data
Integrated Multi-scale	Couple thermal, mechanical, and metallurgical phenomena	Multi-physics coupled models, Digital Twin, AI-assisted simulation	Comprehensive prediction of process, microstructure, and mechanical properties	Improved prediction accuracy and supports real-time optimization	Complex implementation and significant computational resources

distortion using finite element analysis (FEA), making it suitable for process optimization at the component level. Meso-scale modeling employs computational fluid dynamics (CFD) and Volume of Fluid (VOF) methods to simulate melt pool dynamics, fluid flow, and solidification behavior, providing detailed insights into bead formation and heat transfer. Micro-scale modeling utilizes phase-field, cellular automata, and crystal plasticity models to predict grain growth, phase transformations, and microstructure evolution, which directly influence material properties. Integrated multi-scale modeling combines thermal, mechanical, and metallurgical simulations with advanced digital twin and artificial intelligence (AI) techniques to achieve comprehensive process prediction and optimization. Although inte-

grated approaches offer improved predictive accuracy and support intelligent manufacturing, they require significant computational resources and complex model implementation.

2.2 Mesoscale melt pool dynamics

Micro-scale modeling in Wire Arc Additive Manufacturing (WAAM) delves into the detailed behavior of the melt pool and the resulting microstructural transformations, which ultimately determine the mechanical performance of the built component. Unlike macro-scale models, which emphasize thermal gradients and stress distributions across the entire part, micro-scale simulations focus on localized phenomena such as fluid flow, phase transitions, grain morphology, and solute distribution [38]. The high thermal input and complex convection patterns within the melt pool cause significant variations in cooling rates and solidification dynamics, which influence grain growth, texture, and porosity [41]. As such, micro-scale models are vital for predicting metallurgical features and tailoring the microstructure to meet specific application requirements.

One of the primary approaches in micro-scale modeling is Computational Fluid Dynamics (CFD), which is employed to simulate the transient behavior of the melt pool during deposition. These models account for heat transfer, surface tension, Marangoni convection, and electromagnetic forces that govern melt pool shape and flow behavior [37]. For example, Yamamoto et al. [35] developed a CFD model that captured melt pool oscillations during WAAM of stainless steel and showed how arc force and surface tension gradients led to turbulent flow patterns that influenced the solidification front. Their study highlighted the importance of understanding fluid flow in avoiding common defects such as lack of fusion and uneven bead geometry.

In addition to melt pool dynamics, phase-field models are increasingly used to simulate the evolution of the microstructure during solidification. These models allow researchers to track the nucleation and growth of dendrites, columnar grains, and equiaxed grains under non-equilibrium thermal conditions [43]. For instance, Wang et al. [39] used a phase-field model to study Ti–6Al–4 V deposition and revealed that varying arc current and travel speed directly influenced the grain growth direction and texture. The results showed a strong correlation between deposition parameters and the development of columnar versus equiaxed microstructures, which, in turn, affect anisotropy in mechanical behavior. Further integration of micro-scale models with experimental validation has enhanced the predictive accuracy of these simulations. High-speed imaging, infrared thermography, and synchrotron X-ray diffraction have been used in tandem with models to monitor melt pool behavior and solidification in real time [36]. For example, Li et al. [40] employed thermographic data to validate their melt pool shape predictions for aluminum alloy deposition and found that real-time feedback could be used to refine simulation parameters such as viscosity, thermal conductivity, and latent heat. This synergy between modeling and experimentation is crucial for refining simulations to reflect actual process conditions. Microstructure prediction also involves modeling grain refinement strategies such as arc oscillation, interlayer cooling, and alloying additions. Li et al. [44] demonstrated that controlled arc oscillation patterns could manipulate the thermal gradient and solidification rate to produce finer grains and improve mechanical properties. Their simulation results agreed with microstructural analysis using EBSD, demonstrating that simulation-guided grain refinement is achievable. Similarly, Koseki et al. [42] modeled the effects of microalloying elements such as boron and titanium

on grain-boundary pinning and segregation during WAAM, showing that even minor compositional adjustments can lead to significant changes in grain morphology and stability.

Advanced models now incorporate multiphysics simulations that combine fluid flow, thermal history, and solidification kinetics into a unified framework. For example, Zhang and He [45] developed a fully coupled thermofluid-solidification model that could predict not only melt pool size and shape but also dendritic arm spacing and grain orientation. This level of detail is vital for applications where mechanical strength, corrosion resistance, or fatigue performance is closely tied to microstructure, such as in aerospace and marine engineering components. Moreover, machine learning is increasingly integrated with microscale models to accelerate the prediction of microstructural features. Jin et al. [46] trained neural networks on simulation datasets and experimental micrographs to rapidly predict grain size and phase distribution based on input process parameters. This hybrid approach drastically reduces computational time while maintaining accuracy, offering a powerful tool for process optimization and materials design in WAAM. In conclusion, micro-scale modeling in WAAM provides a comprehensive understanding of melt pool behavior and microstructural evolution, enabling the design of materials with tailored properties. These models serve as a bridge between process parameters and material performance, offering predictive insights into grain morphology, phase transformations, and defect formation. As computational capabilities continue to advance, micro-scale simulations are expected to play an increasingly central role in WAAM research, particularly when combined with experimental validation and data-driven techniques for real-time process control.

2.3 Mesoscale melt pool dynamics

2.3.1 Continuity equation

$$\nabla \cdot u = 0$$

Where.

u = velocity vector.

2.4 Navier–stokes equation

$$\rho \left(\frac{\partial u}{\partial t} + u \cdot \nabla u \right) = -\nabla p + \mu \nabla^2 u + F$$

where.

- p = pressure.
- μ = dynamic viscosity.
- F = body forces.

2.5 Marangoni shear stress

$$\tau_M = \frac{\partial \gamma}{\partial T} \nabla_s T$$

where.

- γ = *surface tension*.
- $\nabla_s T$ = *surface temperature gradient*.

Surface tension gradients caused by temperature differences generate Marangoni convection, which strongly influences melt pool circulation, bead geometry, and defect formation.

3 Microscale modeling and microstructure evolution

Microscale modeling in Wire Arc Additive Manufacturing (WAAM) is a crucial approach for understanding and predicting microstructural evolution, particularly grain structure formation and solidification dynamics, and their relationship to mechanical properties. One of the most prominent methods used in this domain is the phase-field method, which enables the simulation of grain growth and phase transitions under transient thermal conditions. The phase-field model excels at capturing grain morphology and evolution during solidification without explicitly tracking the solid-liquid interface, making it particularly suitable for modeling complex dendritic structures in WAAM processes [43]. In their study, phase-field simulations revealed how the imposed thermal gradient and solidification velocity determine whether columnar, equiaxed, or mixed grain structures develop, directly influencing strength and toughness.

A critical feature of solidification modeling involves understanding the solidification front and grain orientation, especially given the high thermal gradients and directional heat dissipation inherent to WAAM. During deposition, steep thermal gradients often lead to epitaxial growth and the formation of elongated columnar grains aligned along the build direction [37]. However, when process parameters are tuned—such as altering travel speed or incorporating interlayer dwell times—equiaxed grains can be promoted, enhancing isotropy in mechanical properties [39]. Simulations that model the movement and curvature of the solidification front enable researchers to predict grain boundary formation and texture evolution, both of which have substantial effects on fatigue life and crack resistance.

In tandem with phase-field methods, cellular automata (CA) have been widely employed to simulate dendritic growth patterns due to their computational efficiency and ability to handle large simulation domains. CA models discretize the solidification space into a lattice of cells, where rules govern the transition of each cell from liquid to solid based on thermal and solutal conditions [47]. These models are particularly effective at predicting grain competition, branching, and coalescence. For example, Raghavan et al. [48] used a coupled CA-phase field model to simulate dendritic growth in Inconel 718. They observed the transition from planar to cellular to dendritic morphologies under different cooling rates. Their model successfully correlated secondary dendrite arm spacing (SDAS) with local solidification

time, an important metric for estimating hardness and yield strength. The insights gained from microscale simulations are further extended by correlating predicted microstructures with mechanical properties, enabling a more predictive design approach. Grain size, shape, and orientation directly affect yield strength, toughness, and ductility through mechanisms such as grain boundary strengthening and texture-induced anisotropy [45]. By simulating the evolution of these features, microscale models provide valuable data for finite element models that incorporate heterogeneous material properties across layers. For instance, Yang et al. [49] demonstrated that microstructure-informed mechanical models could predict local variations in elastic modulus and thermal expansion, which are critical for residual stress prediction and distortion control.

Moreover, microscale models contribute significantly to process optimization by identifying how subtle changes in input parameters influence microstructure. Jin et al. [46] used machine learning-assisted phase-field modeling to create process–microstructure maps for Ti–6Al–4 V, showing how varying current, wire feed rate, and cooling intervals lead to optimized grain refinement strategies. This predictive capability can be particularly useful for designing WAAM components in the aerospace and marine industries, where tailored microstructures are essential to achieve targeted mechanical performance and reliability. In advanced studies, multi-scale coupling is employed, in which microscale models feed data into meso- or macroscale simulations to provide a holistic view of the build process. For instance, grain orientation and distribution obtained from microscale simulations can inform anisotropic constitutive models in thermo-mechanical simulations, thereby enhancing the accuracy of stress and distortion predictions [41]. This integration underscores the role of microscale modeling not only as a standalone tool for microstructural analysis but also as a foundational element of comprehensive WAAM simulation frameworks. In conclusion, microscale modeling plays a vital role in unraveling the complexities of microstructure evolution in WAAM. Through phase-field simulations, cellular automata, and grain growth models, researchers can accurately predict solidification behavior and grain morphology. These predictions form the basis for correlating process parameters with mechanical properties, thereby enabling performance-driven optimization of WAAM processes. As computational power and algorithmic capabilities continue to grow, microscale models will become increasingly precise and integral to the development of next-generation additive manufacturing strategies.

3.1 Micro-scale modelling

3.1.1 Phase-field equation

$$\frac{\partial \phi}{\partial t} = -L\frac{\delta F}{\delta \phi}$$

where.

- ϕ = phase-field variable.
- L = kinetic coefficient.
- F = free energy functional.

Phase-field models simulate grain nucleation, dendritic growth, and phase transformation during solidification without explicitly tracking moving interfaces.

4 Coupled thermal–mechanical–metallurgical modeling

Wire Arc Additive Manufacturing (WAAM) involves complex interactions between heat transfer, mechanical deformation, and microstructural evolution. Since thermal gradients generated during deposition simultaneously influence residual stress development, distortion, phase transformation, and grain morphology, independent modeling approaches are often insufficient for accurate prediction. Therefore, coupled thermal–mechanical–metallurgical modeling has emerged as an essential strategy for developing comprehensive multi-scale simulation frameworks capable of predicting the complete WAAM process chain from heat input to final component properties.

4.1 Strategy for multi-physics coupling

In coupled WAAM simulations, thermal, mechanical, and metallurgical phenomena are integrated through sequential or fully coupled approaches. The thermal model generally acts as the primary component because the predicted temperature field provides the thermal history required for subsequent mechanical and microstructural analysis. The heat transfer equation considers energy input from the arc source, conduction through deposited layers and substrate, convection, and radiation losses. The resulting temperature distribution is then transferred to mechanical models to calculate thermal expansion, plastic deformation, residual stress, and distortion.

Ding et al. [23] emphasized that accurate prediction of WAAM component integrity requires consideration of the interaction between heat accumulation, thermal cycling, and mechanical response during layer deposition. Their review highlighted that thermal history controls not only distortion but also microstructural characteristics and mechanical performance. At the mechanical scale, thermo-mechanical models incorporate temperature-dependent material properties, including elastic modulus, thermal conductivity, yield strength, and thermal expansion coefficient. The calculated thermal strains are used to determine stress evolution during heating and cooling cycles. Hamilton et al. [21] demonstrated that thermal gradients during additive manufacturing generate significant phase transformations and stress variations, highlighting the need for coupling thermal and mechanical analyses.

The metallurgical component provides information regarding phase transformation, solidification behavior, and microstructure evolution. Thermal histories obtained from macro- and mesoscale models are transferred to phase-field or cellular automata models to predict grain nucleation, growth direction, and crystallographic texture. Chen et al. [43] showed that phase-field approaches can successfully simulate solidification behavior under non-equilibrium thermal conditions, providing important links between processing parameters and microstructural characteristics.

A complete multi-scale coupling framework therefore follows the sequence:

$$\textit{Process parameters} \rightarrow \textit{Heat generation} \rightarrow \textit{Melt pool dynamics} \rightarrow \textit{Thermal history} \rightarrow \textit{Stress evolution} \rightarrow \textit{Microstructure development} \rightarrow \textit{Mechanical properties}$$

Such integration allows WAAM models to move beyond isolated predictions and provides a holistic understanding of process–structure–property relationships. Recent developments have focused on fully coupled thermo-fluid-metallurgical models, where melt pool fluid flow, heat transfer, solidification kinetics, and grain growth are solved simultaneously. Zhang and He [45] developed a coupled thermo-fluid-solidification framework capable of predicting melt pool geometry, dendritic growth, and grain orientation, demonstrating the potential of integrated multi-physics simulations.

4.2 Time stepping and convergence challenges

The transient nature of WAAM deposition creates significant computational challenges in coupled simulations. Unlike conventional manufacturing processes, WAAM involves repeated thermal cycles caused by sequential layer deposition, resulting in continuously changing temperature fields and material states. Therefore, appropriate time-stepping strategies are required to capture rapid heating during arc interaction and slower cooling during interlayer periods.

In finite element-based thermal–mechanical simulations, the time increment must be sufficiently small to capture temperature variations and avoid numerical instability. However, extremely small time steps increase computational cost, particularly for large components containing hundreds or thousands of deposited layers. Ding et al. [23] reported that computational efficiency remains one of the major limitations of large-scale WAAM simulations. The coupling between thermal, mechanical, and metallurgical models further increases numerical complexity. Temperature changes affect mechanical properties, while stress development influences deformation and thermal contact conditions. Similarly, phase transformations modify material properties and introduce additional strain components. These interactions require iterative solution strategies to achieve convergence.

Corbin et al. [8] demonstrated that variations in deposition parameters significantly influence thermal gradients and distortion behavior, indicating the importance of accurate transient simulation. Adaptive time-stepping approaches, reduced-order models, and parallel computing strategies are increasingly being explored to overcome computational limitations. Machine-learning-assisted approaches have also been introduced to accelerate coupled simulations. Jin et al. [46] demonstrated that neural networks trained using simulation and experimental datasets can rapidly predict microstructural characteristics, reducing the computational requirements of traditional physics-based models.

4.3 Importance of boundary and initial conditions

Accurate definition of initial and boundary conditions is critical for achieving reliable predictions in coupled WAAM simulations. Since WAAM components are built layer by layer, the initial thermal state, substrate conditions, heat dissipation mechanisms, and interlayer temperature significantly influence the final simulation results.

The substrate temperature, preheating conditions, and ambient cooling environment strongly affect thermal gradients and residual stress development. Sequeira Almeida and Williams [6] demonstrated that controlling substrate conditions and thermal management strategies can significantly reduce stress accumulation during additive manufacturing. Heat source representation is another critical factor influencing simulation accuracy. Gaussian

and double-ellipsoidal heat source models are commonly used to represent arc energy distribution. Incorrect estimation of heat input, efficiency, or heat source geometry can lead to significant differences between predicted and experimental melt pool dimensions. Boundary conditions related to convection, radiation, mechanical constraints, and interlayer contact must also be carefully defined. Xu et al. [12] demonstrated that integrating experimental monitoring data with numerical simulations improves model reliability by providing realistic boundary information during deposition.

Furthermore, material properties used in simulations must account for temperature dependency and phase changes. Simplified assumptions of constant thermal and mechanical properties can introduce errors, particularly in high-temperature regions near the melt pool. Therefore, accurate databases containing temperature-dependent properties are essential for reliable multi-physics modeling. Overall, coupled thermal–mechanical–metallurgical modeling represents the foundation of advanced WAAM simulation frameworks. By integrating heat transfer, stress evolution, and microstructural transformation, these models provide deeper insight into process behavior and enable predictive control of component quality. Future developments combining high-performance computing, reduced-order modeling, digital twins, and artificial intelligence will further improve the efficiency and accuracy of integrated WAAM simulations.

Figure 2 illustrates the integrated thermo–mechanical–metallurgical modeling workflow employed in Wire Arc Additive Manufacturing (WAAM). The process begins with defining key process parameters, including current, voltage, wire feed speed (WFS), and travel speed, which are incorporated into a heat source model, such as the Gaussian or Goldak model. The heat source drives finite element thermal simulations to predict temperature distribution, cooling rate, and thermal gradients. These thermal results are subsequently coupled with thermo-mechanical simulations to estimate residual stresses, distortion, and plastic deformation. Finally, metallurgical simulations predict grain growth and phase transformations, enabling accurate prediction of the resulting mechanical properties, including strength, hardness, fatigue resistance, and overall structural reliability.

5 Model validation and experimental techniques

The accuracy and reliability of multi-scale computational models in Wire Arc Additive Manufacturing (WAAM) depend strongly on appropriate experimental validation. Since WAAM involves complex interactions between heat transfer, fluid flow, phase transformation, residual stress development, and microstructural evolution, numerical predictions must

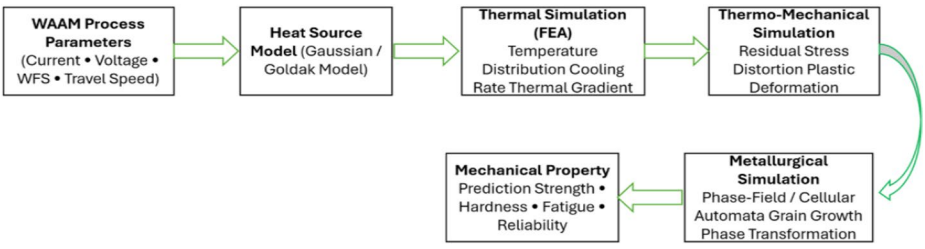


Fig. 2 Coupled thermo–mechanical–metallurgical modeling workflow used for predicting thermal history, stress evolution, and microstructural development in WAAM

be compared with experimentally measured outcomes. Validation provides confidence that computational models can accurately represent physical deposition processes and can be applied for process optimization and defect mitigation. Various experimental techniques, including thermal monitoring, residual stress measurement, and microstructural characterization, are commonly employed to validate macro-, meso-, and micro-scale WAAM models.

5.1 Thermal measurement and thermocouples

Thermal history is one of the most critical factors controlling WAAM process behavior because temperature evolution directly influences melt pool stability, cooling rate, residual stress formation, and microstructural development. Therefore, accurate measurement of transient temperature profiles is essential for validating thermal and thermo-mechanical simulations.

Thermocouples are widely used for measuring temperature evolution during WAAM deposition. In computational models, transient temperature fields are predicted by solving heat transfer equations that account for heat input, material properties, convection, radiation, and boundary conditions. The predicted temperature profiles are then compared with experimentally recorded thermocouple data to evaluate model accuracy. Ribeiro et al. [22] performed one of the early investigations on thermal behavior during gas metal arc-based additive manufacturing. They demonstrated that heat accumulation and cooling conditions strongly influence deposited geometry and process stability. Similarly, Zhang et al. [20] incorporated experimental temperature measurements into weld deposition modeling and showed that thermal monitoring improves the reliability of numerical predictions. More recently, Xu et al. [12] developed a multi-sensor monitoring framework for Ti–6Al–4 V WAAM components by combining temperature measurements with process monitoring techniques. Their study demonstrated that experimentally acquired thermal data can be integrated with numerical models to improve prediction capability and support real-time process control. However, thermocouple-based validation has certain limitations. Due to the highly localized and rapidly changing nature of WAAM heat sources, thermocouples may not capture the extremely high temperature gradients inside the melt pool. In addition, measurement uncertainty can arise due to sensor response time, positioning errors, and repeated thermal cycling. Therefore, thermocouples are increasingly combined with infrared thermography and numerical data assimilation approaches to obtain more reliable thermal validation.

5.2 XRD and residual stress evaluation

Residual stress prediction is a major objective of thermo-mechanical modeling in WAAM because high thermal gradients generated during deposition produce non-uniform expansion and contraction. These stresses can cause distortion, cracking, dimensional inaccuracies, and reduced mechanical performance.

X-ray diffraction (XRD) is one of the most widely used non-destructive techniques for experimental residual stress measurement. The technique evaluates changes in crystal lattice spacing caused by elastic deformation and provides quantitative stress information that can be compared with finite element predictions. Ding et al. [23] highlighted that residual stress

accumulation is one of the major challenges limiting the industrial adoption of WAAM, particularly for large-scale components. Their review emphasized the importance of coupled thermal-mechanical simulations for predicting stress evolution during layer-by-layer deposition. Corbin et al. [8] investigated the relationship between deposition parameters and distortion behavior in additive manufacturing and demonstrated that thermal gradients strongly influence residual stress development. Their findings highlighted the need to validate numerical stress predictions through experimental measurements. For WAAM simulations, validation is generally performed by comparing predicted and measured residual stress profiles along deposition directions, layer interfaces, and different locations within the manufactured component. Quantitative evaluation can be performed using statistical parameters such as root mean square error (RMSE), percentage deviation, and correlation coefficient between simulated and experimental values. Although XRD provides reliable residual stress measurements, differences between numerical and experimental results may occur due to uncertainties in heat source representation, material properties, boundary conditions, and stress relaxation mechanisms. Therefore, accurate calibration of thermal and mechanical models is essential for achieving reliable predictions.

5.3 EBSD and metallographic validation

Microstructural validation is essential in multi-scale WAAM modeling because grain morphology, crystallographic texture, phase distribution, and solidification behavior directly influence mechanical properties. Electron backscatter diffraction (EBSD) has become an important characterization technique for validating phase-field, cellular automata, and other microstructure prediction models.

EBSD provides detailed information regarding grain size, grain orientation, grain boundary characteristics, and texture evolution. These experimentally measured features can be compared with computational predictions to determine whether numerical models accurately represent solidification phenomena. Wang et al. [39] investigated the influence of WAAM process parameters on grain growth and texture evolution in Ti-based alloys. They demonstrated that variations in thermal gradients and cooling conditions significantly affect columnar and equiaxed grain formation. Such experimentally observed microstructural features provide important validation data for micro-scale simulations. McAndrew et al. [15] demonstrated that inter-pass rolling modifies the thermal history and promotes microstructural refinement in Ti–6Al–4 V WAAM components. Their experimental analysis confirmed that changes in processing conditions influence grain morphology and mechanical performance.

Phase-field models are increasingly being validated using EBSD because they can predict grain nucleation, growth direction, and texture development under different thermal conditions. Chen et al. [43] demonstrated the capability of phase-field approaches for predicting solidification structures under non-equilibrium conditions, while Zhang and He [45] further developed coupled thermo-fluid-solidification models capable of predicting melt pool behavior and grain characteristics. Furthermore, Jin et al. [46] highlighted the potential of machine-learning-assisted microstructure prediction by using simulation datasets and experimental microstructural information to estimate grain characteristics rapidly. Such hybrid simulation approaches provide new opportunities for accelerating validation and reducing computational cost.

5.4 Comparison of predicted and actual outcomes

The final stage of model validation involves systematic comparison between simulated results and experimentally obtained outcomes. A reliable WAAM multi-scale model should accurately predict thermal history, melt pool geometry, residual stresses, distortion behavior, and microstructural characteristics. For thermal models, validation is commonly performed by comparing predicted temperature-time profiles with thermocouple or infrared measurements. The accuracy of melt pool simulations is evaluated by comparing predicted bead width, penetration depth, and melt pool shape with experimentally deposited tracks. Thermo-mechanical models are validated by comparing predicted distortion with experimentally measured deformation using techniques such as XRD, digital image correlation, and coordinate measurement systems. Similarly, microstructural models are evaluated by comparing predicted grain size, morphology, and crystallographic texture with EBSD observations.

Xu et al. [12] demonstrated that integrating experimental monitoring with computational models significantly improves WAAM prediction capability and supports the development of digital twin-based manufacturing systems. Similarly, Jin et al. [46] showed that combining simulation-generated datasets with experimental microstructural information can enhance prediction efficiency. Despite significant advances, achieving complete validation of multi-scale WAAM models remains challenging due to uncertainties associated with material properties, dynamic arc behavior, process disturbances, and computational assumptions. Future research should focus on standardized validation methodologies, high-fidelity experimental datasets, uncertainty quantification, and physics-informed artificial intelligence approaches. Overall, experimental validation using thermocouples, XRD, EBSD, and metallographic analysis provides the foundation for developing reliable multi-scale WAAM simulation frameworks. These approaches enable the transition of computational models from theoretical prediction tools toward practical digital manufacturing systems capable of improving quality, reducing defects, and optimizing WAAM processes.

6 Digital twins and AI-augmented simulation

The increasing complexity of Wire Arc Additive Manufacturing (WAAM) processes has created a demand for advanced computational frameworks capable of providing real-time monitoring, prediction, and process optimization. Traditional numerical models based on finite element analysis (FEA), computational fluid dynamics (CFD), and phase-field approaches provide a detailed understanding of thermal behavior, melt pool dynamics, and microstructural evolution; however, their high computational cost limits their direct application for real-time manufacturing control. Digital twin technology combined with artificial intelligence (AI)-augmented simulation has emerged as a promising approach to overcome these limitations by integrating physics-based models, sensor data, and data-driven prediction algorithms into a unified intelligent manufacturing framework.

6.1 Concept of real-time modeling in WAAM

A digital twin can be defined as a dynamic virtual representation of a physical WAAM process that continuously exchanges information with the actual manufacturing system through sensors and data acquisition platforms. Unlike conventional offline simulations, digital twins update their predictions in real time using experimental measurements obtained during deposition.

In WAAM, a digital twin framework typically integrates multiple information sources, including arc current, voltage, wire feed rate, travel speed, temperature distribution, melt pool geometry, and component deformation. Computational models process these real-time inputs to predict process evolution and provide feedback for adaptive manufacturing control. Xu et al. [12] developed a multi-sensor monitoring framework for Ti–6Al–4 V WAAM components and demonstrated that combining experimental monitoring with computational models improves process understanding and enables advanced control strategies. Their study highlighted the importance of integrating sensor information with numerical prediction frameworks for future intelligent manufacturing systems.

The digital twin concept extends traditional simulation by creating a continuous connection between the physical and virtual environments:

$$\begin{aligned} &\textit{Physical WAAM system} \rightarrow \textit{Sensors} \rightarrow \textit{Data acquisition} \\ &\rightarrow \textit{Digital model} \rightarrow \textit{Prediction} \rightarrow \textit{Process adjustment} \end{aligned}$$

Such frameworks allow manufacturers to predict defects before they occur, optimize deposition parameters, and reduce dependence on trial-and-error experimentation. Nilsiam et al. [13] demonstrated the potential of open-source GMAW-based metal additive manufacturing systems by integrating monitoring and control capabilities, highlighting the importance of accessible digital platforms for future WAAM development. However, implementing real-time digital twins remains challenging because high-fidelity simulations often require significant computational resources. Therefore, reduced-order models, surrogate models, and artificial intelligence techniques are increasingly being incorporated to enable faster prediction.

6.2 Training machine learning models using simulation datasets

Machine learning (ML) has emerged as an effective approach for accelerating WAAM simulations by learning complex relationships between process parameters, thermal behavior, microstructure, and final properties. Since experimental data generation in WAAM is expensive and time-consuming, simulation-generated datasets provide valuable sources for training predictive models.

Physics-based simulations can generate large datasets containing information about:

1. Heat distribution.
2. Melt pool dimensions.
3. Cooling rates.
4. Residual stress evolution.
5. Grain morphology.
6. Mechanical properties.

These datasets can then be used to train machine-learning algorithms such as artificial neural networks (ANNs), support vector machines (SVMs), random forests, and deep learning models.

Jin et al. [46] demonstrated the potential of machine-learning-assisted microstructure prediction by training neural networks using simulation outputs and experimental microstructural datasets. Their approach significantly reduced computational time while maintaining prediction accuracy for grain characteristics and phase distribution. Similarly, data-driven approaches can establish process–structure–property relationships by identifying patterns that are difficult to capture using conventional analytical models. For example, ML models can predict the influence of wire feed rate, arc current, and travel speed on melt pool geometry and microstructural evolution without performing computationally expensive simulations for every parameter combination.

The combination of physics-based simulations and machine learning provides a hybrid modeling strategy. In this approach, physical models provide scientific understanding and boundary constraints, while ML algorithms improve computational efficiency and prediction speed. Nevertheless, the accuracy of ML-based models strongly depends on the quality and diversity of training datasets. Limited experimental data, inconsistent processing conditions, and differences between simulation assumptions and real manufacturing environments remain important challenges.

6.3 Predictive control and anomaly detection

One of the major advantages of digital twins and AI-assisted simulation is their ability to enable predictive process control and early detection of manufacturing abnormalities. In conventional WAAM systems, defects such as porosity, excessive bead irregularity, distortion, and lack of fusion are often detected after fabrication. AI-based monitoring systems aim to identify these problems during deposition and automatically adjust process parameters.

Predictive control systems continuously analyze sensor data and compare real-time measurements with model predictions. If deviations occur, the system can modify parameters such as current, voltage, wire feed rate, or travel speed to maintain optimal deposition conditions. Hu et al. [14] investigated abnormal bead formation in GMAW-based additive manufacturing and demonstrated the importance of process monitoring for controlling deposition defects. Their findings showed that variations in thermal behavior and bead geometry can be identified through real-time monitoring approaches. Digital twin-based control systems can also support anomaly detection by identifying unusual patterns in thermal signals, melt pool characteristics, or mechanical responses. Machine-learning algorithms trained on normal operating conditions can recognize deviations associated with defects and provide early warnings.

Xu et al. [12] emphasized that multi-sensor frameworks combined with computational models provide the foundation for intelligent WAAM platforms capable of monitoring manufacturing progress and improving process reliability. Future developments are expected to integrate physics-informed neural networks (PINNs), reinforcement learning, and autonomous control algorithms with digital twin architectures. These approaches will enable WAAM systems to learn from previous manufacturing cycles and continuously improve process performance. Overall, digital twins and AI-augmented simulations represent a major advancement toward intelligent WAAM manufacturing. By combining real-time sensing,

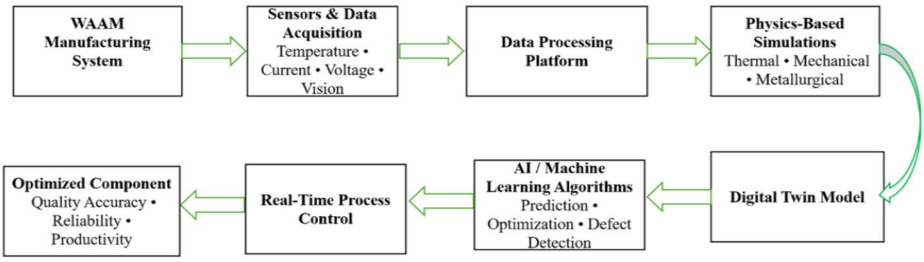


Fig. 3 Digital twin and AI-enabled framework for real-time monitoring, prediction, optimization, and process control in WAAM

physics-based modeling, and machine learning, these technologies provide opportunities for autonomous process optimization, defect prevention, and improved component reliability.

Figure 3 illustrates the integration of digital twin technology and artificial intelligence (AI) for intelligent Wire Arc Additive Manufacturing (WAAM). The framework begins with the WAAM manufacturing system, where sensors continuously collect process data, including temperature, current, voltage, and visual information. These data are processed and integrated with physics-based simulations encompassing thermal, mechanical, and metallurgical models to create a digital twin of the manufacturing process. AI and machine learning algorithms analyze the simulated and real-time data to predict process behavior, optimize manufacturing parameters, and detect defects. The generated insights enable real-time process control, resulting in enhanced component quality, improved dimensional accuracy, increased reliability, and higher manufacturing productivity.

7 ~~Case studies~~ and industrial applications

The transition of Wire Arc Additive Manufacturing (WAAM) from laboratory-scale research to industrial production requires reliable computational tools capable of predicting process behavior, minimizing defects, and ensuring structural integrity. Large-scale metallic components manufactured by WAAM are increasingly being considered for aerospace, marine, defense, and energy applications due to their high deposition rate, material efficiency, and capability to produce complex geometries. However, industrial implementation remains challenging because large components experience severe thermal accumulation, residual stress development, distortion, and microstructural variations. Multi-scale modeling approaches provide an effective strategy for addressing these challenges by enabling virtual process optimization before actual fabrication.

7.1 ~~Large~~ aerospace structural part simulation

Aerospace industries represent one of the most promising application areas for WAAM due to the requirement for large, lightweight, and high-strength metallic structures. Conventional manufacturing of aerospace components often involves extensive machining from large billets, resulting in significant material waste and high production costs. WAAM offers an alternative approach by producing near-net-shape components with reduced material consumption.

However, manufacturing large aerospace structures through WAAM requires accurate prediction of thermal accumulation, distortion, and mechanical performance. Multi-scale simulations are therefore employed to optimize deposition strategies, predict residual stresses, and improve dimensional accuracy. Ding et al. [23] highlighted that large WAAM components experience complex thermal histories due to repeated heating and cooling cycles during layer deposition. Their review emphasized that numerical modeling is essential for controlling distortion and maintaining structural reliability in aerospace-scale components. Xu et al. [12] demonstrated the application of a multi-sensor monitoring framework for Ti–6Al–4 V WAAM components, integrating experimental measurements with computational predictions. Their approach provided valuable information regarding temperature evolution, deposition progress, and process stability, supporting the development of digital manufacturing strategies.

For aerospace alloys such as Ti–6Al–4 V and Inconel-based materials, coupled thermal–mechanical models are frequently used to simulate temperature distribution, stress evolution, and deformation during fabrication. Corbin et al. [8] demonstrated that deposition parameters strongly influence thermal gradients and mechanical distortion in additive manufacturing components, emphasizing the importance of simulation-driven parameter selection.

In industrial aerospace applications, multi-scale modeling enables:

1. Optimization of deposition path planning.
2. Prediction of component distortion before manufacturing.
3. Reduction of machining allowance.
4. Control of thermal history for improved microstructure.
5. Development of defect-free manufacturing strategies.

The integration of finite element models with microstructure prediction techniques further improves reliability by linking process parameters with final mechanical properties.

7.2 WAAM repair scenarios for marine applications

Marine industries have shown significant interest in WAAM for repair and refurbishment of large metallic components such as ship structures, propellers, valves, and offshore equipment. Unlike conventional replacement methods, WAAM repair allows damaged regions to be restored directly, reducing downtime and material costs. However, repair applications introduce additional challenges because deposition occurs on previously manufactured or service-exposed components. Differences in substrate temperature, geometry, residual stress state, and material condition can strongly affect bonding quality and repair performance. McAndrew et al. [15] demonstrated that inter-pass rolling during Ti–6Al–4 V WAAM modifies thermal history and promotes microstructural refinement, improving mechanical performance. Such approaches are particularly valuable in repair applications where controlling residual stress and restoring mechanical properties are critical.

The use of multi-scale modeling allows engineers to predict:

1. Heat-affected zone development.
2. Residual stress accumulation during repair.
3. Interface bonding quality.
4. Microstructural compatibility between deposited and existing material.

For marine applications, corrosion resistance and fatigue performance are also important considerations. Microstructure-based simulations provide insight into grain morphology and texture development, which directly influence long-term service behavior. Zhang and He [45] demonstrated that coupled thermo-fluid-solidification models can predict melt pool characteristics and microstructural evolution, providing a pathway for improving repair quality through simulation-based parameter optimization. Furthermore, digital twin concepts are increasingly being explored for repair operations, where real-time sensor monitoring data are combined with numerical models to evaluate repair progress and predict final component performance.

7.3 Optimization for minimal residual stress

Residual stress remains one of the most significant barriers to the industrial adoption of WAAM because large thermal gradients generate tensile stresses that may lead to distortion, cracking, and premature failure. Therefore, optimization strategies aimed at minimizing residual stress are a major focus of WAAM modeling research.

Thermal–mechanical simulations allow researchers to evaluate the influence of process parameters such as:

1. Arc current.
2. Travel speed.
3. Wire feed rate.
4. Deposition sequence.
5. Interlayer cooling time.
6. Substrate preheating.

before performing physical experiments.

Ding et al. [23] reported that controlling heat accumulation and deposition strategies is essential for reducing residual stress in large WAAM components. Similarly, Corbin et al. [8] showed that variations in deposition parameters significantly affect stress distribution and final distortion behavior.

Inter-pass techniques have also been investigated as strategies for stress reduction. McAndrew et al. [15] demonstrated that inter-layer rolling can modify thermal and mechanical conditions, resulting in microstructural refinement and improved component performance. Advanced optimization approaches increasingly combine numerical simulations with artificial intelligence. Machine-learning models trained using simulation datasets can rapidly predict stress distribution and identify optimal processing conditions without requiring extensive computational simulations. Jin et al. [46] demonstrated the potential of AI-assisted prediction approaches for accelerating material and process optimization.

Future industrial WAAM systems are expected to combine multi-scale simulation, digital twins, and AI-based optimization to enable autonomous manufacturing. Such integrated frameworks will allow real-time adjustment of process parameters to minimize residual stress, improve dimensional accuracy, and enhance component reliability. Overall, case studies from aerospace and marine applications demonstrate that multi-scale modeling is not only a theoretical research tool but also a practical approach for enabling industrial-scale WAAM adoption. Through accurate prediction of thermal behavior, stress evolution,

and microstructural development, simulation-driven manufacturing can significantly reduce experimental trials and improve the reliability of large metallic components.

8 Future trends and research opportunities

The future development of Wire Arc Additive Manufacturing (WAAM) depends on improving the accuracy, efficiency, and scalability of computational modeling approaches. Although significant progress has been achieved in thermal–mechanical simulation, melt pool modeling, microstructure prediction, and digital twin development, several challenges remain before fully autonomous and predictive WAAM systems can be realized. The major research directions include improved multi-scale integration, cloud-based simulation platforms, and high-performance computing strategies for accelerating complex simulations.

8.1 Multiscale integration improvements

Multi-scale modeling has become an essential approach for understanding the complex relationships between WAAM process parameters, thermal behavior, microstructural evolution, and mechanical performance. However, current modeling approaches often treat macro-scale thermal–mechanical behavior, mesoscale melt pool dynamics, and microscale microstructure evolution as separate problems. Future research should focus on developing fully integrated multi-scale frameworks where information can be continuously exchanged between different length and time scales.

At the macro-scale, finite element models provide predictions of temperature distribution, residual stress, and distortion during large-scale deposition. At the mesoscale, computational fluid dynamics (CFD) models describe melt pool geometry, fluid flow, surface tension effects, and solidification behavior. At the microscale, phase-field and cellular automata approaches predict grain formation, texture development, and phase transformation. The challenge is to establish efficient coupling strategies that transfer information between these models without high computational cost. Ding et al. [23] highlighted that accurate prediction of WAAM component performance requires considering interactions between thermal history, stress development, and microstructural evolution rather than evaluating individual phenomena independently. Similarly, Zhang and He [45] demonstrated that coupled thermo-fluid-solidification models provide improved prediction capability by integrating melt pool behavior with microstructure development.

Future research should focus on developing adaptive multi-scale models where the required level of simulation detail changes according to process requirements. For example, high-resolution simulations may be applied only near critical regions such as melt pool boundaries, defect locations, or stress concentration zones. At the same time, reduced models can be used for less sensitive regions. Such approaches will improve both accuracy and computational efficiency. Another important research direction is integrating physics-based models with artificial intelligence. Hybrid physics-informed machine learning approaches can maintain scientific accuracy while improving prediction speed. Jin et al. (2022) [46] demonstrated that machine-learning-assisted microstructure prediction can significantly reduce computational time by learning relationships between simulation parameters and material characteristics.

8.2 Cloud-based simulation frameworks

The increasing complexity of WAAM simulations requires significant computational resources, particularly for large industrial components involving thousands of deposition layers. Cloud-based simulation frameworks provide a potential solution by enabling researchers and industries to access high-performance computational resources without maintaining expensive local infrastructure.

Cloud platforms can support collaborative simulation environments where experimental data, numerical models, and machine-learning algorithms are integrated into a common digital framework. Such systems are particularly beneficial for industries requiring rapid design optimization and remote manufacturing decision-making. Digital twin-based WAAM systems can greatly benefit from cloud computing because real-time sensor data generated during manufacturing can be transferred to cloud platforms for analysis and prediction. Xu et al. [12] demonstrated the importance of sensor-based monitoring frameworks for WAAM process control, providing a foundation for future cloud-connected manufacturing systems.

Cloud-based simulation also enables large-scale data storage and sharing, which is important for developing machine-learning models. Since WAAM processes generate large datasets involving thermal histories, melt pool characteristics, and microstructural information, cloud environments can facilitate the development of standardized databases for AI-assisted manufacturing. However, several challenges must be addressed before widespread implementation of cloud-based WAAM simulation. These include data security, communication latency, model interoperability, and the requirement for standardized simulation formats. Future research should focus on developing secure and efficient cloud architectures specifically designed for additive manufacturing applications.

8.3 Computational efficiency and high-performance computing

Computational efficiency remains one of the greatest challenges in multi-scale WAAM simulation. High-fidelity models involving coupled thermal, mechanical, fluid flow, and metallurgical phenomena require significant computational time due to complex geometries, nonlinear material behavior, and transient deposition conditions.

High-performance computing (HPC) approaches provide opportunities to overcome these limitations by enabling parallel processing, distributed computing, and accelerated numerical solutions. Large-scale WAAM simulations can benefit from GPU-based computing, parallel finite element solvers, and advanced numerical algorithms. Reduced-order modeling (ROM) has also emerged as a promising approach for improving simulation efficiency. Instead of solving complete high-dimensional models, reduced-order approaches identify dominant physical behaviors and create simplified models that provide rapid predictions. These approaches are particularly valuable for real-time process control and digital twin applications.

Xu et al. [12] emphasized that real-time monitoring and prediction require computational frameworks capable of rapidly processing manufacturing data. Therefore, combining HPC with reduced-order models and artificial intelligence will be essential for achieving real-time WAAM optimization. Artificial intelligence-based surrogate models represent another important research direction. Machine-learning algorithms trained using high-fidelity simulation datasets can provide rapid predictions of thermal behavior, residual stress, and micro-

structural characteristics. Jin et al. [46] demonstrated that AI-assisted prediction approaches can significantly accelerate microstructure modeling while maintaining acceptable accuracy.

8.3.1 Future HPC-based WAAM frameworks are expected to integrate

1. Parallel multi-physics simulation.
2. GPU acceleration.
3. Reduced-order modeling.
4. Physics-informed neural networks.
5. Cloud-based computational resources.
6. Real-time digital twin platforms.

These developments will enable faster design optimization and support autonomous manufacturing systems.

Overall, future research in WAAM modeling will focus on creating integrated, intelligent, and computationally efficient simulation frameworks. Improvements in multi-scale coupling, cloud-based platforms, and high-performance computing will allow WAAM to transition from experimental development toward fully predictive and autonomous manufacturing. By combining physics-based simulation with artificial intelligence and advanced computational infrastructure, future WAAM systems will achieve improved reliability, reduced defects, and enhanced industrial scalability.

9 Conclusion

Multi-scale modeling provides a comprehensive and effective framework for understanding, predicting, and optimizing Wire Arc Additive Manufacturing (WAAM) processes. Since WAAM involves complex interactions among heat transfer, melt pool behavior, stress development, and microstructural evolution, a single-scale approach is often insufficient. The integration of finite element analysis, thermal–mechanical simulations, and phase-field modeling enables accurate prediction of temperature distribution, residual stresses, distortion, grain growth, and solidification behavior. Macro-scale models support the evaluation of global heat flow and structural deformation, while mesoscale models provide valuable insights into melt pool geometry, bead formation, and dilution. At the microscale, phase-field and related methods help explain microstructural transformations that directly affect mechanical performance.

Coupling thermal, mechanical, and metallurgical models, supported by experimental validation through thermocouples, XRD, and EBSD, improves the reliability of simulation outcomes. Advanced approaches such as cellular automata, reduced-order modeling, digital twins, and data-driven techniques further enhance computational efficiency and real-time process control. Overall, multi-scale modeling reduces dependence on trial-and-error experimentation, improves dimensional accuracy, minimizes defects, and supports sustainable manufacturing. Therefore, simulation-driven WAAM design is essential for producing high-quality large-scale metallic components in aerospace, defense, marine, and other advanced engineering applications.

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Declarations

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